K-Nearest Neighbors Algorithm in Machine Learning

Introduction

K-Nearest Neighbors (KNN) is a simple yet powerful **supervised learning algorithm** used for both classification and regression. It makes predictions based on the majority class (or average) of the nearest data points in the feature space.

How it Works

1. Choose the number of neighbors K.
2. Compute the distance (e.g., Euclidean, Manhattan) between the test point and training points.
3. Select the K *nearest neighbors.*
4. For classification: assign the class with the majority vote.
5. For regression: return the average value of neighbors.

Example

If $K=3$ and the nearest neighbors of a point are two "Diabetic" and one "Non-Diabetic", then the point is classified as "Diabetic".

Applications

- **Medical Diagnosis**: Predicting diseases based on patient history.
- **Recommendation Systems**: Suggesting products based on user similarity.
- **Pattern Recognition**: Handwriting or face recognition.
- **Anomaly Detection**: Detecting fraud in banking.

Advantages

- Easy to implement, no training phase.
- Works well with small datasets.
- Flexible for both classification and regression.

Limitations

- Computationally expensive with large datasets.
- Sensitive to irrelevant or scaled features.
- Choosing the right K *is crucial: small K ->* noisy, large K *->biased.*

Improvements

- Use of **KD-Trees or Ball Trees** for faster searches.
- Weighted KNN: closer neighbors have more influence.
- Feature scaling (Normalization/Standardization) is essential.